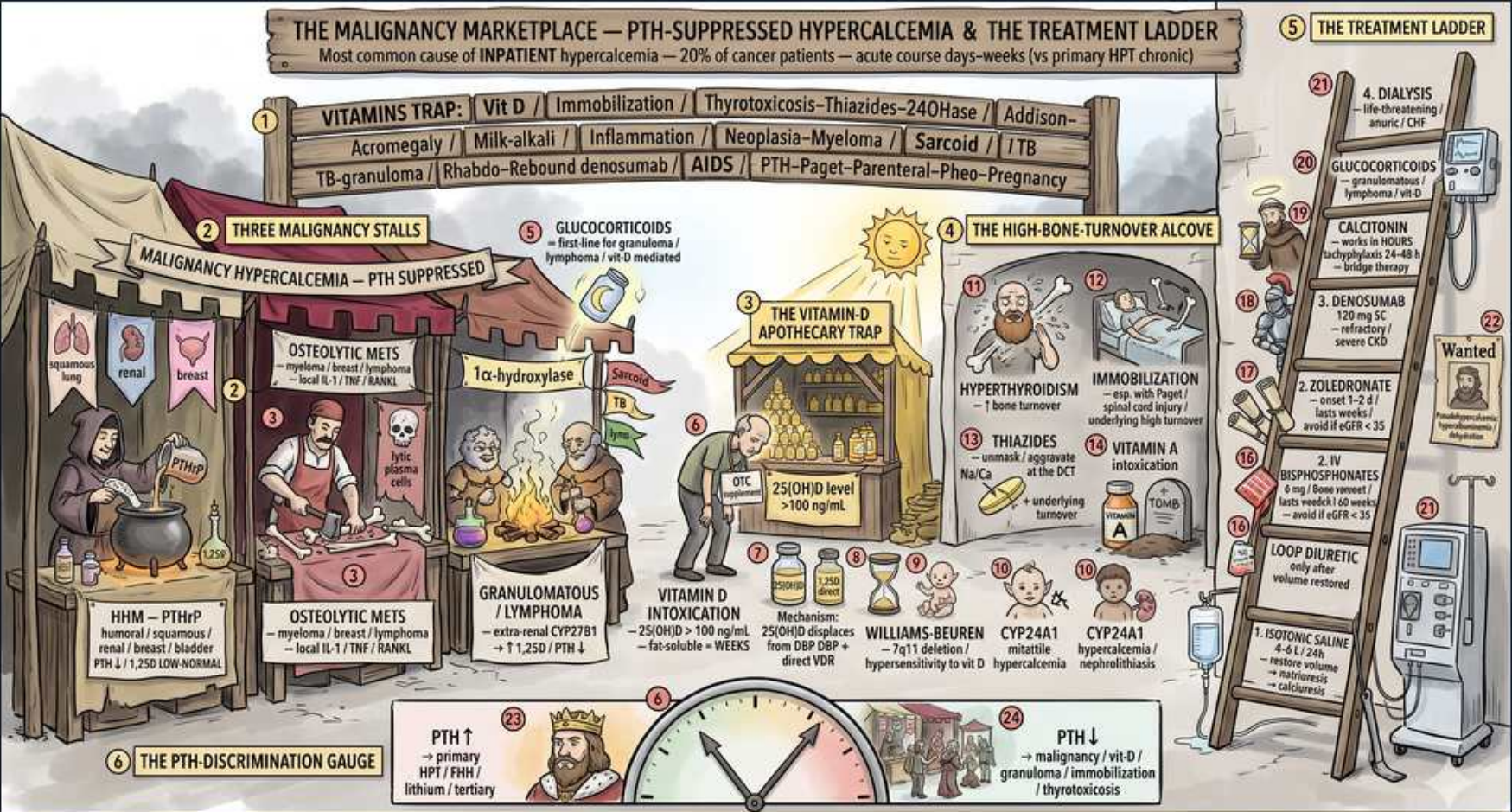


Hypercalcemia of Malignancy — Mechanisms & Treatment Ladder



Inpatient acute hypercalcemia → think malignancy. Granuloma / lymphoma → high 1,25D + LOW 25(OH)D + glucocorticoid-responsive. Vit D intox → high 25(OH)D. SALINE FIRST, then bisphosphonate. Calcitonin = bridge only.

1. Malignancy = #1 inpatient hyperCa

WHAT YOU SEE

Banner: most common cause of inpatient hypercalcemia

WHAT IT ENCODES

Malignancy is the leading cause of INPATIENT hypercalcemia (about 20% of cancer patients) and presents acutely over days to weeks with PTH SUPPRESSED. Primary hyperparathyroidism is the leading OUTPATIENT cause and is chronic.

BOARD TRAP

Wrong answer: primary hyperparathyroidism. Discriminator: malignancy is acute/symptomatic with LOW PTH; primary HPT is chronic/mild with HIGH PTH.

2. PTHrP — humoral hypercalcemia (HHM)

WHAT YOU SEE

HHM = PTHrP stall

WHAT IT ENCODES

Humoral hypercalcemia of malignancy: tumor secretes PTHrP (squamous cell lung/head-neck, renal, bladder, breast). PTHrP mimics PTH → high Ca, LOW phosphate, but PTH suppressed and 1,25-vitamin D usually low/normal.

BOARD TRAP

Wrong answer: bone metastases. Discriminator: HHM is humoral (PTHrP) with FEW or no bone mets; measure PTHrP.

3. Osteolytic metastases / local cytokines

WHAT YOU SEE

Osteolytic mets stall, RANKL

WHAT IT ENCODES

Local osteolytic hypercalcemia: bone mets (breast, multiple myeloma) release local cytokines and RANKL driving osteoclastic bone resorption. Myeloma uses RANKL/MIP-1alpha; PTH and PTHrP are suppressed.

BOARD TRAP

Wrong answer: PTHrP-mediated HHM. Discriminator: osteolytic type has extensive bone mets and normal PTHrP; myeloma shows M-protein and lytic lesions.

4. Granulomatous / lymphoma — 1,25-vit D

WHAT YOU SEE

Granulomatous / lymphoma stall, 1,25 high

WHAT IT ENCODES

Macrophages/granulomas (sarcoid, TB) and some lymphomas express extrarenal 1-alpha-hydroxylase → high 1,25(OH)₂D → increased gut Ca absorption. Hypercalcemia is glucocorticoid-RESPONSIVE.

BOARD TRAP

Wrong answer: dietary vitamin D excess (high 25-OHD). Discriminator: granulomatous disease raises 1,25-D specifically and responds to steroids.

5. Vitamin D intoxication — high 25-OHD

WHAT YOU SEE

Vitamin D apothecary trap, 25-OHD >150

WHAT IT ENCODES

Exogenous vitamin D toxicity raises 25(OH)D (often >150) with high Ca and high phosphate. Fat-soluble, so hypercalcemia lasts weeks. Distinguish from granulomatous disease which raises 1,25-D, not 25-D.

BOARD TRAP

Wrong answer: granulomatous 1,25-D excess. Discriminator: intoxication shows markedly elevated 25(OH)D; granuloma elevates 1,25(OH)₂D.

6. PTH discrimination gauge

WHAT YOU SEE

PTH dial: primary vs malignancy/vit D

WHAT IT ENCODES

First fork in any hypercalcemia: check PTH. HIGH/inappropriately-normal PTH = primary/tertiary hyperparathyroidism or FHH. LOW PTH = malignancy, vitamin D, granuloma, hyperthyroidism, immobilization.

BOARD TRAP

Wrong answer: jumping to malignancy workup before PTH. Discriminator: PTH FIRST — low PTH then pursue PTHrP/vitamin D metabolites.

7. Hyperthyroidism — increased bone turnover

WHAT YOU SEE

High-bone-turnover alcove, thyroid

WHAT IT ENCODES

Thyrotoxicosis raises bone turnover and can cause mild hypercalcemia with suppressed PTH. Treat the underlying hyperthyroidism. Part of the PTH-independent (low-PTH) hypercalcemia group.

BOARD TRAP

Wrong answer: malignancy. Discriminator: hyperthyroid hypercalcemia is mild, with thyrotoxic signs and suppressed TSH.

8. Immobilization

WHAT YOU SEE

Immobilization figure with cast

WHAT IT ENCODES

Prolonged immobilization (especially in young patients or Paget disease) causes hypercalcemia from increased osteoclastic resorption with low PTH. Classic in spinal cord injury or extensive casting.

BOARD TRAP

Wrong answer: occult malignancy. Discriminator: immobilization hypercalcemia follows immobility and resolves with mobilization/bisphosphonate.

9. Thiazides — distal Ca reabsorption

WHAT YOU SEE

Thiazides label, DCT

WHAT IT ENCODES

Thiazides increase distal tubular calcium reabsorption and can unmask/worsen hypercalcemia. Stop thiazides as part of management. Often a contributor rather than sole cause.

BOARD TRAP

Wrong answer: loop diuretic effect. Discriminator: thiazides RAISE Ca (distal reabsorption); loops LOWER Ca (block paracellular reabsorption).

10. Step 1: IV saline volume repletion

WHAT YOU SEE

Treatment ladder top: saline first

WHAT IT ENCODES

First-line acute therapy is aggressive isotonic SALINE to restore volume and promote calciuresis. Corrects the dehydration that perpetuates hypercalcemia. Add loop diuretic only AFTER euvolemia if volume overload.

BOARD TRAP

Wrong answer: routine furosemide up front. Discriminator: saline FIRST; loops are no longer routine and only after volume repletion.

11. Calcitonin — fast bridge

WHAT YOU SEE

Calcitonin step, fast onset

WHAT IT ENCODES

Calcitonin lowers calcium within hours (inhibits osteoclasts, increases renal excretion) but tachyphylaxis limits it to ~48 hours. Used as a BRIDGE while bisphosphonate takes effect.

BOARD TRAP

Wrong answer: calcitonin as definitive therapy. Discriminator: calcitonin is a short bridge (tachyphylaxis); bisphosphonate provides durable control.

12. Bisphosphonate — definitive (zoledronate)

WHAT YOU SEE

Zoledronate / pamidronate step

WHAT IT ENCODES

IV bisphosphonate (zoledronic acid) is definitive for malignancy hypercalcemia; onset 2-4 days, lasts weeks via osteoclast inhibition. Dose-adjust/avoid in severe renal impairment; watch osteonecrosis of jaw.

BOARD TRAP

Wrong answer: expect immediate effect. Discriminator: bisphosphonate takes 2-4 days — pair with saline+calcitonin for the acute period.

13. Denosumab — RANKL inhibitor

WHAT YOU SEE

Denosumab step, RANKL antibody

WHAT IT ENCODES

Denosumab (anti-RANKL) is used for bisphosphonate-refractory hypercalcemia or severe renal impairment (not cleared by kidney). Risk of hypocalcemia. Effective in myeloma/osteolytic disease.

BOARD TRAP

Wrong answer: avoid in renal failure. Discriminator: denosumab is PREFERRED over bisphosphonate when GFR is very low (no renal clearance).

14. Glucocorticoids — vit D-mediated only

WHAT YOU SEE

Glucocorticoids step, lymphoma/granuloma

WHAT IT ENCODES

Glucocorticoids work for 1,25-D-mediated hypercalcemia: granulomatous disease, lymphoma, and vitamin D intoxication. They reduce extrarenal 1-alpha-hydroxylase activity and gut Ca absorption.

BOARD TRAP

Wrong answer: steroids for PTHrP/osteolytic hypercalcemia. Discriminator: glucocorticoids only help VITAMIN D-mediated (granuloma/lymphoma) types.
